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Report Highlights:

Norway is the biggest producer of fish in Europe and the number one seafood exporter in the world. In 2024, 2.26 MMT of salmonid fish feed was used with a total estimated value of $3.16 billion for the ingredients, of which about $2.80 billion were imported. All plant protein ingredients are non-genetically modified and predominantly imported from Brazil. Most of the fishmeal and fish oil is sourced from the Nordic European region and South America. A part of the fish oil ($75 million in 2024) is sourced from the United States. Besides a brief overview of the Norwegian fish feed market, this report includes also the Norwegian import requirements of the most used ingredients.

Summary

Norway is the biggest producer of fish in Europe and the number one seafood exporter in the world. The main family group of fish species cultured is the Salmonids for which about 2.26 million metric tons (MMT) of fish feed was used in 2024. Despite the inclusion of vegetable ingredients in the feeds having increased from about 25 percent in 2000 to nearly 70 percent in 2024, marine ingredients are still indispensable, and in terms of value, the most important components. The feed ingredients have a total estimated value of $3.16 billion with fish oil ($723 million), fishmeal ($603 million), soybean protein concentrates ($492 million), rapeseed oil ($487 million), and wheat gluten ($315 million) representing the highest value.

In 2024, Norway imported $2.80 billion of fish feed ingredients in addition to $330 million of soybeans and soybean meal for livestock and poultry feeds. Given the progress made in fattening carnivorous fish with soy derivatives, the Norwegian fish feed market can be regarded as the innovative edge of the soy feed market. All plant proteins are non-genetically modified (non-GM) and predominantly imported from Brazil. Most of the fishmeal and oil is sourced from the Nordic European region (Northeast Atlantic) and South America (Southeast Pacific). A part of the fish oil ($75 million in 2024, nine percent of total Norwegian fish oil imports) is sourced from the United States.

Besides a brief overview of the Norwegian fish feed market, this report includes the Norwegian import requirements of the most used fish feed ingredients. Feed products, including additives, which consist of, contain or are produced based on genetically modified organisms (GMOs), are not permitted to be sold or marketed unless the Norwegian Food Safety Authority has approved this. Despite the fact that GM fish feed ingredients are not used in Norway, Seafood Norway is open to the use of plant products from GMOs in feed for farmed species on the terms set by the authorities.

The Norwegian Fish Feed Market

Aquaculture in Norway – The Raw Materials as Input for the Blue Revolution

During the past fifty years, Norway has shown leadership in the so-called "blue revolution," which refers to the global expansion of aquaculture. Currently, Norway is the biggest producer of fish in Europe and produces about a third of the output of the whole European Union (see table 1). In Norway, the main family group of fish species cultured is the Salmonids with a live weight volume of 1.63 MMT and a primary production value of $11.55 billion in 2023. Within this group, the main species cultured is the Atlantic salmon with a volume of 1.54 MMT and a primary production value of $10.94 billion in 2023 (source: Eurostat). To produce salmonid fish species the Norwegian aquaculture sector used 2.26 MMT of fish feed in 2024 (see table 2).

Table 1. Main European Fish Producers (1,000 MT)

	2016	2017	2018	2019	2020	2021	2022	2023
Greece	100	106	110	105	112	130	130	123
Spain	68	71	73	76	65	70	78	84
Italy	54	57	50	54	48	60	48	51

Poland	35	35	37	40	45	40	42	41
France	47	46	43	46	47	44	41	40
Croatia	17	16	19	19	21	27	26	26
Denmark	33	32	29	34	32	29	28	24
EU Total	*518*	*535*	*529*	*539*	*539*	*569*	*563*	*553*
United Kingdom*	162	190	156	204	192	205	169	151
Norway	*1,324*	*1,306*	*1,353*	*1,451*	*1,488*	*1,663*	*1,659*	*1,648*

Source: Eurostat *Atlantic Salmon production in Scotland which represents about 95 percent of the UK fish production.

Table 2. Raw Materials Use for Salmonid Feed in Norway (1,000 MT)

	2000[ab]	**2010[ab]**	**2020[ab]**	**2024[b]**
Fishmeal	207	334	318	385
Fish Oil	189	227	254	294
Vegetable Meal	134	481	869	1,087
Vegetable Oils	0	174	424	430
Other	79	120	254	68
Total	**610**	**1,337**	**2,120**	**2,264**

Sources: Aquaculture Reports: (a) *Utilization of feed resources in the production of Atlantic salmon (Salmo salar) in Norway: An update for 2020* and (b) Salmon Farming Handbook 2025.

Important components of the feed are marine fishmeal and fish oil. One of the main challenges of the sector is the scarcity of these materials, as well as the high fluctuation in availability and quality. In 2009, the Norwegian Government's strategy for an environmentally sustainable Norwegian aquaculture industry stated that the raw materials for fish feed shall originate from sustainably managed stocks, thereby avoiding further depletion of marine resources. Following this strategy and driven by economic reasons, the industry has been substituting marine ingredients with vegetable proteins and oils. Due to the efforts of fish nutritionists, the inclusion of fishmeal and fish oil declined from about 65 percent in 2000 to 30 percent in 2024 (see table 2). In Norway, feed materials of plant origin currently make up for about 67 percent of the feed for salmonids, equivalent to a volume of roughly 1.52 MMT. About 70 percent of the vegetable components consist of proteins and carbohydrates (1.87 MMT), and about 30 percent of vegetable oils (430,000 MT).

Vegetable Ingredients – The Innovative Edge

Looking more closely into the use of the specific ingredients, soybean protein concentrate (SPC) is the most extensively used plant ingredient in salmonid fish feeds in Norway (see table 3). SPC represents an estimated volume of 473,000 MT with a market value of about $492 million. SPC falls under HS code 23.09.90.40 (fish feeds). In 2024, total Norwegian imports of fish feeds were $633 million. Given the progress made in fattening carnivorous fish with soy derivatives, the Norwegian fish feed market can be regarded as the innovative edge of the soy feed market. Other plant ingredients used at a lower inclusion rate are gluten (from wheat and corn) and bean and seed proteins (from guar beans, sunflower seeds, and peas). SPC has the benefit of a relatively high protein content (about 65 percent).

In 2000, the use of vegetable oils in salmonid feeds was negligible but increased strongly up to 2020. In 2024, rapeseed oil was by far the most used vegetable oil. In 2024, Norway imported 374,000 MT of rapeseed oil for fish feeds with a value of $385 million from Belarus, Russia, France, Belgium, and Denmark. The most used vegetable carbohydrate is wheat, which is predominantly imported from Germany, Poland and Sweden.

Despite the fact that GM fish feed ingredients are not used in Norway, Seafood Norway, representing the aquaculture private sector, is open to the use of plant products from GMOs in feed for farmed species on the terms set by the authorities. The main reason for sourcing non-GM is because the Norwegian aquaculture sector is highly export focused. Norway is the largest fish and seafood exporter in the world, with a market value of $15.4 billion in 2024. In 2021, the Norwegian Seafood Council (NSC), the marketing organization working together with the Norwegian seafood industry, announced it would only source from deforestation-free supply chains. Together with the sustainability owner ProTerra and World Wide Fund for Nature (WWF) Brazil, the soy suppliers have agreed on a monitoring, reporting and verification system to implement and enforce their commitment to zero deforestation. According NSC, Norwegian salmon feed is already completely non-GM, free from antibiotics, and are marine ingredients sourced from certifiably sustainable sources and strictly controlled for any unwanted substances. SPC used in Norway's salmon feed is primarily supplied by Brazilian producers.

Table 3. Ingredients of Salmonid Feed in Norway[a]
(Volume in 1,000 MT, Unit Value in $/MT, and Value in million $)

Ingredient	HS Code	Percentage in 2020	Volume in 2024	Unit Value in 2024[b]	Value in 2024 (imported)
Vegetable Proteins		**40.5**	**917**	**-**	**1,200 (992[c])**
-Soy Protein Concentrate	23.09.90.40	20.9	473	1,040	492
-Wheat Gluten	11.09.00.10	9.8	222	1,420	315
-Guar Protein	13.02.32	4.3	97	2,765	269
-Sunflower Protein	21.06.10.02	3.4	77	975	75
-Pea Protein	21.06.10.02	1.4	32	975	31
-Corn Gluten	23.03.10.13	0.7	16	1,100	17
Vegetable Oils		**20.2**	**457**	**-**	**561 (405)**
-Rapeseed Oil	15.14.11	18.0	408	1,195	487
-Linseed Oil	15.15.11	1.3	29	1,270	37
-Soybean Oil	15.07.90	0.4	9	1,600	14
-Camelina Oil	15.15.90	0.4	9	2,000	18
-Coconut Oil	15.13.19	0.1	2	2,000	5
Vegetable Carbohydrates		**12.6**	**285**	**-**	**103 (159[c])**
-Wheat	10.01.99.01	6.5	147	300	44
-Faba Beans	07.13.50.11	3.6	82	470	38
-Pea Flour	07.13.10.01	2.5	57	360	20
Fishmeal		**12.1**	**274**	**-**	**603 (440[c])**

-Meal of Forage Fish	23.01.20.10	8.8	199	2,200	438
-Meal of Trimmings	23.01.20.10	3.3	75	2,200	164
Fish Oil		**10.3**	**233**	**-**	**723 (828[c])**
-Oil of Forage Fish	15.04.20.11	8.3	188	3,100	583
-Oil of Trimmings	15.04.20.11	2.0	45	3,100	140
Micro-ingredients	**-**	**4.5**	**102**	**-**	**-**
Total	**-**	**100**	**2,264**	**-**	**3,190**

(a) Based on the feed ration of Atlantic salmon in Norway in 2020 and volume of salmonid feed use in 2024. Sources: Salmon Farming Handbook 2025, Aquaculture Reports: *Utilization of feed resources in the production of Atlantic salmon (Salmo salar) in Norway: An update for 2020* (b) Source: Trade Data Monitor, value estimated by FAS The Hague (c) includes feed for other animals.

Imports of Soybeans and Meal for the Livestock and Poultry Sector

Norway annually imports about 350,000 MT of soybeans and 150,000 MT of soybean meal, with a value of respectively $195 million and $135 million in 2024. The majority of the soybeans and meal are imported from Brazil, and, to a lesser extent from Canada and the European Union (Romania and Poland). Prior to the introduction of GM soy crops around the year 2000, the United States supplied about 40 percent of Norway's soy. Denofa, the only soy crusher in Scandinavia, processes the imported soybeans, producing about 280,000 MT of meal and about 70,000 MT of oil (based on average European conversion factors). Taking Norwegian imports and export in account, about 250,000 MT of the soybean meal is available for the domestic Norwegian compound feed market, of which is mainly for the cattle, swine, and poultry sectors. Most of the domestically produced soybean oil is exported to Sweden and the Netherlands.

Fishmeal and Oil – Still Indispensable

The inclusion of plant materials is bound to a range of limitations mainly related to the presence of anti-nutritional factors negatively affecting the palatability and digestibility of the feed. Another important factor is the deficiency of nutrients such as amino acids and fatty acids, which are both essential for the growth performance of the fish and the wholesomeness of the fish filet. Despite the progress made to replace marine ingredients with plant proteins and oils, the use of fishmeal and oil continued to grow because of its indispensability and the production volume of fish feed sustained its growth since 2000 (see table 2).

In Norway, the use of fishmeal and fish oil to produce salmonid fish feed is estimated at respectively 274,000 MT and 233,000 MT, with an estimated value of $603 million and $723 million (see table 3). Most of the meal and oil is produced from forage fish and a smaller share from fish trimmings. In 2024, Norway imported 218,000 MT of fishmeal for animal feeds with a value of $440 million. Most of the meal is imported from Iceland, Denmark, the Faroe Islands, and Ireland and produced from fish caught in the Northeast Atlantic. A smaller portion is imported from Uruguay, Chile, and Peru and produced from fish caught in the Southeast Pacific and Antarctic Atlantic.

In 2024, Norway imported 155,000 MT of fish oil for animal feed with a value of $828 million. Similar as reported for the source of the meal, the largest share of the fish oil is produced from fish caught in the

Northeast Atlantic and imported from Denmark and Iceland. The second largest share is produced from fish caught in the Western-central Atlantic (including Gulf of America). Other important sources are the Southeast Pacific and the Eastern Central Pacific. Fish oil produced from fish caught in the three latter fishing areas is imported from Chile, Mexico, the United States, Panama, and Peru. In 2024, Norway imported 14,600 MT of fish oil from the United States with a value of $75 million.

Norwegian Legislation and Import Requirements for Fish Feed Ingredients

Vegetable Ingredients

For feed materials of plant-origin, such as plant protein concentrates and gluten, imported into Norway from third countries (i.e., outside the EU/European Economic Area (EEA)) the key regulatory authority is the Norwegian Food Safety Authority (known in Norwegian as Mattilsynet).

- Feed regulation is contained in the Norwegian Regulation No. 1290 of 7 November 2002, Regulation on Feed Products (*Forskrift om fôrvarer*).
    - Appendix I contains provisions on thresholds for contaminants (regulating for instance the maximum levels for arsenic, cadmium, fluorine, lead, mercury, nitrite, and melamine, aflatoxin, and other unwanted substances). The regulation also contains provisions on microbes, and more specifically § 14. stipulates specific requirements in relation to Salmonella.
- Feed products, including additives, which consist of, contain or are produced on the basis of genetically modified organisms (GMOs), are not permitted to be sold or marketed unless the Norwegian Food Safety Authority has approved this. In Norway, no genetically modified products have yet been approved for use as food, feed for terrestrial animals or as seed products. Therefore, there should be no such products on the market. For fish feed, one product has been approved that can be freely imported and traded. It is Aquaterra® oil from a rapeseed that has been genetically modified to provide an increased content of several long-chain omega-3 fatty acids. Since products using Aquaterra rapeseed are not yet approved in the EU, feed mixtures containing this rapeseed oil cannot be exported to the EU. Despite its approval, the product is not used in Norway.
- The Norwegian regulation for import controls is Regulation No. 717 of March 9, 2020 On Official Controls – Import Controls of Non-Animal Origin Products (*Forskrift om offentlig kontroll – importkontroll av ikke animalske produkter*), which is an implementation of EU Regulation 2019/1793.
- All feed materials of plant-origin destined for Norway, must be notified to the Norwegian Food Safety Authority (*Mattilsynet*) through an approved system at least 24 hours before arrival. To do this, a Common Health Entry Document (CHED) is required.
- Feed hygiene regulations apply. For example, Regulation (EC) 183/2005 has become part of Norwegian regulation as Regulations on feed hygiene (*Forskrift om fôrhygiene*).
- For labeling the Regulation on the Labelling and Marketing of Feed Products (*Forskrift om merking og omsetning av fôrvarer*) applies.
- There is a strong emphasis on certification, traceability, and responsible sourcing of fish feed ingredients. The Norwegian Seafood Industry Association, mentions that soy as feed needs to be certified by the Round Table on Responsible Soy (RTRS) standard or the ProTerra standard, and that feed manufacturers "are bound by open industry standards set out by the European Feed Manufacturers Federation (FEFAC), which are based on global certification standards."

Fishmeal and Fish Oil

- Fishmeal and fish oil fall under Norwegian Regulation No. 1064 of September 14, 2016 Regulations on Animal By-Products that are not Intended for Human Consumption ("The Animal By-Products Regulation") (*Forskrift om animalske biprodukter som ikke er beregnet på konsum (aninaliebiproduktforskriften)*).
- The Norwegian Regulation on official controls (*Forskrift om offentlig kontroll*) implements EU Regulation 2021/405, which lists third countries or regions from which certain animal products may be imported into the EEA.
- Exporting fish meal and fish oil to Norway is only allowed after a facility is correctly registered as processing plant in the EU's Processing Plant Establishment List (ABP-FSB). This can be done by registering through the National Oceanic and Atmospheric Administration (NOAA), who outlines the validation process in Chapter 6 of its Seafood Inspection Manual. Once this process is completed, it is possible to obtain export certificates through NOAA's Seafood Inspection Program (SIP) online certificate system, which has dedicated a webpage to the import requirements for Norway.
- All fish meal and fish oil destined for Norway, must be notified to the Norwegian Food Safety Authority (*Mattilsynet*) through an approved system at least 24 hours before arrival. Each consignment must be pre-notified in TRACES. To do this, a Common Health Entry Document (CHED) must be created.
- Norway has five veterinary border control posts, handling specific categories of products. For detailed information, check the most up to date version on the website of Mattilsynet.
- The import fees for fish meal and complete feed for fish are currently NOK 25.75 ($2.55) per MT, and NOK 21.42 ($2.12)if intended for marine mammals, see the Norwegian Regulation on payments of fees for veterinary border control.
- Feed hygiene regulations apply. For example Regulation (EC) 183/2005 has become part of Norwegian regulation as Regulations on feed hygiene (*Forskrift om fôrhygiene*).
- For labeling the Regulation on the Labelling and Marketing of Feed Products (*Forskrift om merking og omsetning av fôrvarer*) applies. Specifically mentioned is that feed containing fishmeal must be labelled with "contains fish, shall not be given to ruminants".
- There is a strong emphasis on certification, traceability, and responsible sourcing of fishmeal and fish oil. The Norwegian Seafood Industry Association, mentions that salmon feed needs to be certified by the IFFO RS standard (International Marine Ingredients Organization Standard).
    - The Norwegian government has set a goal, that by 2034, all fish feed will come from sustainable sources, as announced in a 2024 press release.

For more information, see also:

- Feed requirements | The Norwegian Food Safety Authority
- Fish Feed Monitoring Program | The Norwegian Food Safety Authority

For more information about the Norwegian fish feed market, please contact FAS The Hague at AgTheHague@usda.gov.

Attachments:

No Attachments.